

Scheme/ Answer Key for Valuation			
Scheme of evaluation (marks in brackets) and answers of problems/key			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY			
B.TECH S2 (R,S) EXAMINATION (2019 SCHEME), JUNE 2022			
Course Code: CYT 100			
Course Name: ENGINEERING CHEMISTRY			
(2019 -Scheme)			
Max. Marks: 100			Duration: 3 Hours
PART A			
		Answer all questions, each carries 3 marks	Marks
1		Determination 3 marks	(3)
2		At equilibrium $E_{\text{cell}} = 0$ and $Q=K$ - 1mark, $0= 1.1-(0.0591/2) \times \log K$ $K=1.67 \times 10^{37}$ 2 marks	(3)
3		Ans c and d 2 marks , explanation or values 1 mark	(3)
4		$3 \times 3 - 5 = 4$ modes 1 marks sketch 2 marks	(3)
5		1 mark each	(3)
6		1.5 mark each	(3)
7		Definition 1 mark illustration keto and enol forms 2 marks	(3)
8		Definition 1 mark poly acetylene 1 mark, poly aniline 1 mark	(3)
9		Total hardness = $300 \times 100/40 + 192 \times 100/24 = 1550$ ppm Temp Hardness= $122 \times 50/61 = 100$ ppm Perm hardness = 1450 ppm 1 mark each	(3)
10		Definition 1 mark Determination 2 marks	(3)
PART B			
Answer one full question from each module, each question carries 14 marks.			
MODULE 1			
11	a	Definition 2 marks, any 5 Applications 1 mark each	(7)
	b	Definition 2 marks equation 1 mark , Cell constant determination 2 marks , Conductivity of sample 2 marks	(7)
12	a	Principle 3 marks , equations 2 marks, two applications 2 marks	(7)
	b	Sacrificial 4 marks impressed current 3 marks	(7)
MODULE 2			
13	a	Principle 5 marks applications 3 marks	(8)
	b	3 marks each	(6)

14	a	Definition 2 marks differential form 1 mark deduction of integrated form 2 marks Graph 2 marks slope = ϵl , 1 mark	(8)
	b	$\mu = \frac{12 \times 16}{12 + 16} \times 1.66^{-27} = 11.38 \times 10^{-27} kg$ 2 marks $k = 4\pi^2 \mu c^2 \overline{\nu^2} = 1849 Nm^{-1}$ 4 marks	(6)
MODULE 3			
15	a	Principle 2 marks Instrumentation 3 marks diagram 2 marks. $CaC_2O_4 \cdot H_2O$ graph 3 marks	(10)
	b	explanation with graph 4 marks	(4)
16	a	Principle 2 marks Procedure 3 marks Visualisation any two 2 marks	(7)
	b	Principle 2 marks Instrumentation 3 marks working 2 marks .	(7)
MODULE 4			
17	a	Definition 2 marks condition 2 mark, Example 3 marks (Appropriate marks may be given if any distinction in physical, chemical or biological properties of enantiomers are answered .)	(7)
	b	view ↓ Newman staggered Front C, 60° anticlock 2 R 3 S Conversion 4 marks assigning R,S - 3 marks	(7)
18	a	Definition 1 marks synthesis (chemical structure or structure shown below) 2 marks, Two properties 2 marks and two application 2 marks	(7)

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	b	Definition 1 mark, synthesis 2 mark, two properties 2 marks, two applications 2 marks.	(7)
MODULE 5			
19	a	Definition 2 marks, structure 1 mark demineralisation 3 marks , regeneration 2marks	(8)
	b	Definition 1 mark explanation 2 marks diagram 2 marks, advantage 1 mark	(6)
20	a	Flow diagram 1 mark, primary 2 marks, secondary 4 marks tertiary 3 marks	(10)
	b	BOD = $(7.4-3.8) \times 6 = 21.6$ ppm - 4 marks	(4)
